

Chapter 11 Replacement and Retention Decisions

- **Reasons for replacing an asset**

- Reduced Performance
 - Wear and tear
 - Decreasing reliability and productivity
 - Increasing operating and maintenance costs
- Altered Requirements
 - New production needs, accuracy, speed, etc.
- Obsolescence
 - Current assets may be less productive
 - Not state of the art – need to meet competition

- **Terminology**

- *Defender Asset*. Currently installed asset.
- *Challenger Asset*. Could replace the defender asset.
- *Equivalent Uniform Annual Cost (EUAC)*. AW for costs.
- *Economic Service Life (ESL)*. Number of years at which lowest AW of costs occur.
- *Defender First Cost*. Initial investment in the defender (P).
- *Market value (MV_k) of the defender in year k* . Salvage value of the defender in year k .
- *Challenger first cost*. Initial cost necessary to acquire and install the challenger.

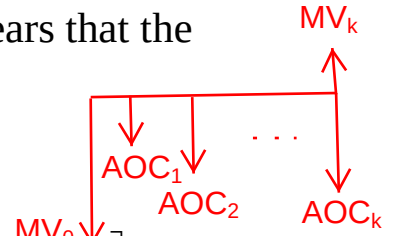
- **Economic Service Life (ESL)**

- Number of years, k^* , at which the equivalent annual worth AW_k of costs is minimized over all possible years that the asset may provide a needed service.

- AW_k is given by

$$AW_k = \boxed{-MV_0(A/P, i, k) + MV_k(A/F, i, k)} - \left[\sum_{j=1}^k AOC_j(P/F, i, j) \right] (A/P, i, k)$$

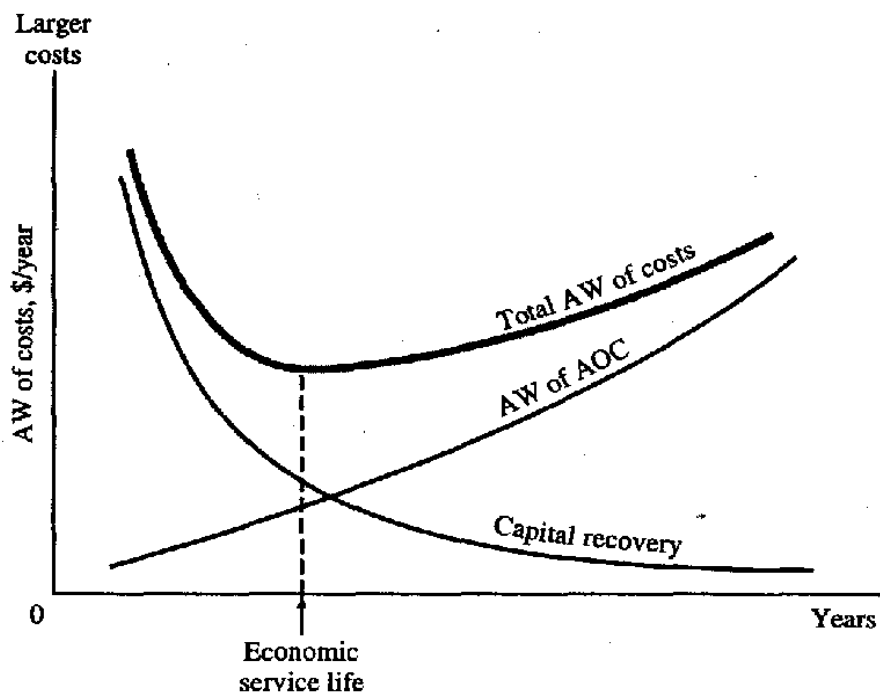
$$= -CR_k - AW_k(AOC)$$



- Symbolically,

$$k^* = \arg \min_k |AW_k| = \arg \min_k CR_k + AW_k(AOC).$$

- Usually, AOC increase with time, k and CR decreases with k .
- The ESL, k^* , strikes a balance between these two costs.
- High CR means that the asset has been utilized long enough and generated enough value.



- **Special cases for ESL**

- If AOC are the same for all years, then k^* . Then, set k^* at the maximum possible number of years.
- If AOC is increasing with time and MV is the same for all years, then replace in year 1. Then, $k^* = 1$.

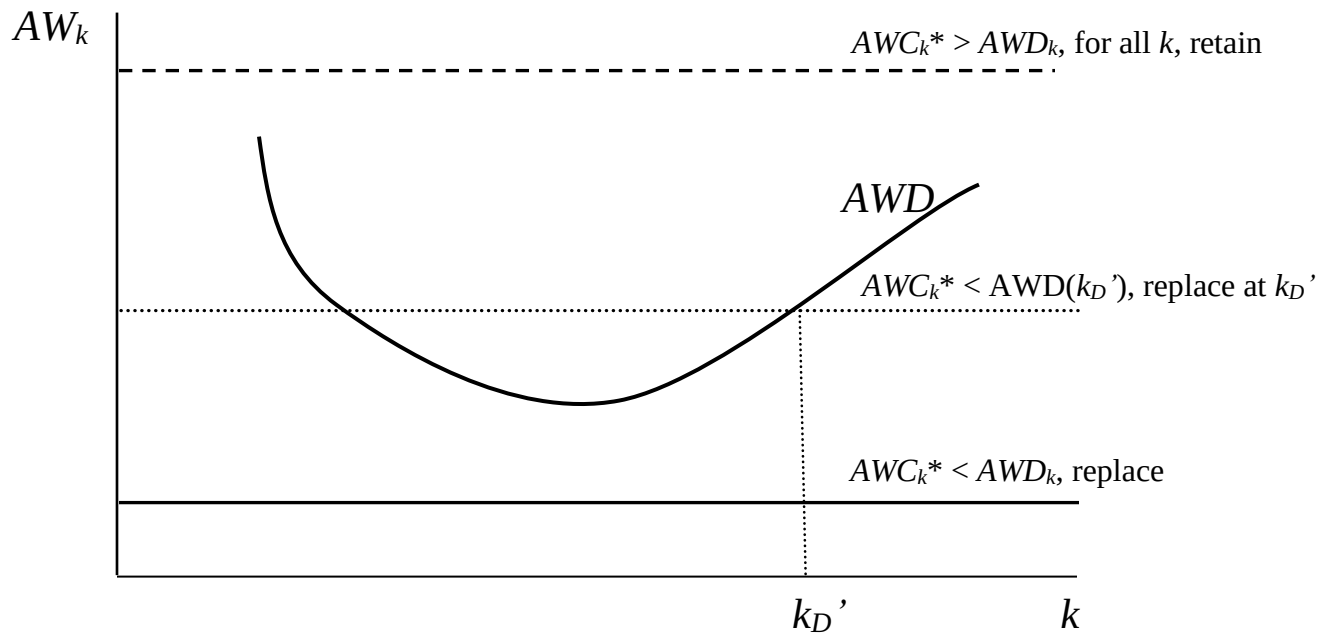
- **Replacement with an identical challenger**

- We are considering here setting a replacement policy, i.e., the planning horizon is assumed to be long (infinity).
- Suppose an asset is to be replaced with an identical challenger. Let k_0 be the age of the defending asset.
- Then, replace the asset if $k_0 \geq k^*$, where k^* is the ESL. Otherwise (if $k_0 < k^*$), then keep the asset.
- This will minimize the long run cost assuming that replacement is only possible with identical assets.
- For example, an asset with $k^* = 3$ should be replaced if its age is 3 or more years, and retained, otherwise.

- **Replacement with a nonidentical asset**

- Suppose we are comparing (now) a defender asset with challenger having different characteristics.
- The implicit assumption is that whichever asset is chosen, it will be replaced each ESL with an identical asset (later).
- Let k_D^* and AWD_k^* be the ESL and corresponding AW of the defender, and k_C^* and AWC_k^* be those of the challenger.

- Then, replace the defender *immediately* if $AWC_k^* < AWD_k^*$.
- Otherwise,
 - If $AWC_k^* > AWD_k^*$, but $AWC_k^* < AWD_k$ for some year $k_D' > k_D^*$, then replace the defender at year k_D' . *If no such k_D' exists, then keep defender.*
 - If $AWC_k^* > AWD_k$, for all k , then keep the defender as long as possible.



- The assumption here is that no other challenger will be identified in future years.

- **Generalized replacement models**

- Sometimes one can expect emergence of future challengers.
- In these cases, one should include future alternatives in the analysis.
- This complicates the replacement problem since many options are now possible.
- For example, suppose there is a challenger now and a challenger next year. The options are:
 - (i) Replace now with current challenger and retain current challenger thereafter.
 - (ii) Replace now with current challenger and replace and replace with next year's challenger.
 - (iii) Retain now and replace with next year's challenger.
 - (iv) Retain defender throughout.